
StadiumMind AI

Product & System Design Package

FIFA World Cup 2026 · Flagship venue: MetLife Stadium

Version 1.0 · July 18, 2026

Confidential — Product, Design & Engineering

Contents

1. Product Requirements Document

2. User Journey Maps

3. Information Architecture

4. Low-Fidelity Wireframes

5. High-Fidelity Design System

6. System Design Document

7. Implementation Roadmap

StadiumMind AI — Product Requirements Document (PRD)

Version: 1.0 · **Status:** Approved for build · **Owner:** Product **Event:** FIFA World Cup 2026 · **Flagship venue:** MetLife Stadium (Final)

1. Executive summary

StadiumMind AI is a GenAI-powered operating system for FIFA World Cup 2026 stadiums. It unifies five audiences — **fans, volunteers, operations managers, security officers and medical teams** — on one real-time decision platform that turns live crowd, transport, weather and incident data into *explained* recommendations a human can approve in one tap.

The World Cup 2026 will be the largest edition ever: **48 teams, 104 matches, 16 host cities across 3 countries, ~6–7 million in-stadium attendees**. Venues must move enormous crowds safely, keep queues short, respond to incidents in seconds, and deliver a world-class fan experience in dozens of languages. Today, that coordination is fragmented across radios, spreadsheets, CCTV walls and disconnected apps.

StadiumMind AI closes that gap with a **multi-agent AI orchestrator** sitting on top of an **event-driven data platform**, surfaced through a **mobile fan app** and role-specific **operations, security and volunteer** consoles.

The one-sentence pitch

See → Tap → AI explains → Human approves. Every stakeholder sees the same live truth, and AI proposes the next best action with a reason attached.

2. Goals & non-goals

2.1 Product goals

1. **Reduce queue and navigation time** for fans by $\geq 30\%$.
2. **Cut incident response time** (detection → responder on scene) by $\geq 40\%$.
3. **Prevent dangerous congestion** through prediction and pre-emptive redirection.
4. **Deliver equitable access** — accessibility-first routing and multilingual assistance.
5. **Give operators calm, explainable control** — never a wall of alarms.

2.2 Business goals

- Be the reference "smart stadium" platform for host-city operators.

- Demonstrable ROI: fewer stewards per 1,000 fans, lower incident liability, higher fan CSAT.
- Extensible to leagues and other mega-events post-tournament.

2.3 Non-goals (v1)

- Ticketing / payments (integrate, don't rebuild).
- Broadcast / replay video products.
- Physical hardware (sensors/turnstiles) — we ingest, we don't manufacture.
- Autonomous action without human approval (explicitly out of scope for safety).

3. Personas (summary)

Persona	Primary need	Core screen
Fan	"Where do I go, and how do I avoid queues?"	Fan mobile app
Volunteer	"Where am I needed right now?"	Volunteer app
Operations Manager	"Keep the venue flowing without drowning in data."	Ops dashboard
Security Officer	"Detect and coordinate incident response fast."	Security console
Medical Team	"Reach the patient by the fastest clear route."	Medical view
Accessibility user	"A dignified, step-free path to everything."	Fan app (a11y mode)
Transport authority	"Meter egress to match transport capacity."	Ops transport panel

Full journey maps: see `01_USER_JOURNEYS.md`.

4. Problem statement & opportunity

Problem. Mega-event venues are coordinated reactively. Congestion is noticed after it forms; incidents are relayed by voice; fans wander; information is siloed by role and language. The result is longer queues, slower response, safety risk and a diminished experience.

Opportunity. Live data (crowd density, transit feeds, weather, incident reports) already exists but is not *fused* or *acted upon* intelligently. GenAI can (a) fuse it, (b) predict where problems will occur, (c) recommend the best action, and (d) explain that action in plain language to each stakeholder in their own language — while keeping a human firmly in control.

5. Product principles

1. **Minimal UI.** Show the next best action, not every datum.
2. **AI assists, never overwhelms.** No alarm storms; ranked, deduplicated recommendations.
3. **Explain before recommending.** Every recommendation carries a reason and a confidence.
4. **Mobile-first & accessibility-first.** Designed for a thumb and for everyone.
5. **Human approval for operational actions.** AI proposes; a person disposes.

6. **Multilingual by default.** Language is a first-class setting, RTL included.
 7. **Fail safe & offline-tolerant.** Degrade gracefully; never hard-crash a live venue.
-

6. Scope — functional requirements

6.1 Fan app (mobile)

- FR-F1 Language selection on first run (6 languages incl. RTL Arabic) and switchable anytime.
- FR-F2 Home: match card, one-tap primary actions, live alerts — no typing required.
- FR-F3 Interactive stadium map (Google Maps satellite + congestion overlay; SVG fallback).
- FR-F4 Wayfinding to seat / food / washroom / exit / accessibility, congestion-aware.
- FR-F5 AI assistant (voice + suggested prompts + free text) with action cards.
- FR-F6 Live alerts and push notifications scoped to the fan's location.
- FR-F7 Accessibility mode: step-free routing, larger targets, screen-reader support.
- FR-F8 Transport & egress guidance after the match, based on live transit load.

6.2 Operations dashboard

- FR-O1 Live stadium heatmap (per-zone congestion).
- FR-O2 Ranked, explained AI recommendations with Approve/Reject (human-in-the-loop).
- FR-O3 Incident timeline and status.
- FR-O4 Volunteer allocation & dispatch view.
- FR-O5 Transport/egress metering panel.
- FR-O6 KPIs: attendance, congested zones, open incidents, match clock.

6.3 Security & medical consoles

- FR-S1 Incident intake (type, severity, zone) and triage.
- FR-S2 Fastest clear-route to incident; corridor protection (fan reroute).
- FR-S3 Cross-team coordination log with timestamps and audit trail.

6.4 Volunteer app

- FR-V1 Current assignment + turn-by-turn to the demand point.
- FR-V2 Accept / en-route / on-scene / resolved status updates.
- FR-V3 Broadcast and targeted messages from Ops.

6.5 AI platform

- FR-AI1 Six domain agents: Crowd, Navigation, Emergency, Accessibility, Transport, Sustainability.
- FR-AI2 Proactive analysis cycle producing ranked recommendations.
- FR-AI3 Reactive intent handling (fan/operator queries).
- FR-AI4 Natural-language explanations via LLM (Gemini), with deterministic fallback.
- FR-AI5 Crowd forecasting (gradient-boosted models) and OR optimisation (queueing, routing, allocation).

6.6 Platform & data

- FR-P1 Real match data (football-data.org) on the fan/ops surfaces.
- FR-P2 Real geolocation of the venue and its gates/exits (Google Maps + Directions).
- FR-P3 Event-driven ingestion of crowd, transit, weather and incident streams.
- FR-P4 Role-based authentication and authorization.

7. Non-functional requirements

Area	Requirement
Performance	Fan app interactive < 2.5 s on 4G; API p95 < 300 ms; heatmap refresh ≤ 2 s.
Scale	90,000 concurrent fans/venue; 16 venues; 500 msg/s per venue on the event bus.
Availability	99.95% during match windows; graceful degradation to cached/last-known state.
Latency (AI)	Recommendation cycle ≤ 3 s end-to-end; LLM explanation ≤ 2 s (async, non-blocking).
Security	OAuth2/OIDC, RBAC, TLS 1.3, at-rest encryption, full audit log of operator actions.
Privacy	No PII in crowd analytics; aggregate density only; GDPR/local compliance per host country.
Accessibility	WCAG 2.1 AA; screen-reader flows; ≥ 44px targets; contrast ≥ 4.5:1.
i18n	6 launch languages incl. RTL; all copy externalised; locale-aware dates/numbers.
Observability	Metrics, tracing, structured logs; per-agent decision logging for audit.
Resilience	Every external dependency has an offline fallback; no single point of failure.

8. Success metrics (KPIs)

North-star: *Fan minutes saved per match* (navigation + queue time avoided).

KPI	Baseline	Target
Avg. queue time at gates	18 min	≤ 12 min
Avg. navigation time to seat	9 min	≤ 6 min
Incident detection → responder on scene	6–8 min	≤ 4 min
Peak congestion events (red zones) per match	—	↓ 50% vs uncontrolled
Recommendation approval rate	—	≥ 70% (proxy for usefulness)
Fan CSAT (in-app)	—	≥ 4.5 / 5
Accessibility route success rate	—	≥ 98%

9. Assumptions & dependencies

- Venue provides crowd-density feeds (CCTV analytics / Wi-Fi/BLE counts) or we simulate them.
- Transit authorities expose arrival/departure feeds (or GTFS-RT).
- Google Maps Platform and a match-data provider (football-data.org) are licensed.
- Gemini (Vertex AI) access for LLM explanations.
- Host-venue Wi-Fi/cellular sufficient for the fan app at peak.

10. Risks & mitigations

Risk	Impact	Likelihood	Mitigation
Connectivity saturation at peak	Fan app unusable	High	Offline-first caching; edge CDN; SMS fallback for critical alerts.
AI over-trust / bad recommendation	Safety/ops error	Med	Human approval mandatory; confidence + explanation; audit trail.
Data feed outage (transit/crowd)	Degraded intelligence	Med	Graceful fallbacks; last-known-good; simulation backstop.
LLM latency/cost spikes	Slow explanations	Med	Async explanations; deterministic template fallback; caching.
Privacy/regulatory scrutiny	Legal blocker	Med	Aggregate-only analytics; DPIA per host country; no face ID.
Multi-venue rollout complexity	Delivery risk	High	Single-tenant-per-venue deploy; config-driven venue model.

11. Release criteria (v1 "match-ready")

- All FR-F/O/AI/P marked "must" pass functional + system tests.
- Load test sustains 90k concurrent fans/venue at target latencies.
- Security review + DPIA signed off.
- Accessibility audit (WCAG 2.1 AA) passed.
- Runbooks + on-call rota for match days; rollback tested.
- Bilingual+ content QA for all 6 launch languages (incl. RTL).

12. Out-of-scope backlog (post-v1)

Digital twin & IoT sensor fusion · drone/robotics dispatch · BLE/UWB indoor positioning · RL-based crowd control · predictive maintenance · sponsor/retail personalization · cross-venue "tournament control tower".

Companion documents: 01_USER_JOURNEYS.md , 02_INFORMATION_ARCHITECTURE.md ,
03_WIREFRAMES.md , 04_DESIGN_SYSTEM.md , 05_SYSTEM_DESIGN.md , 06_ROADMAP.md .

StadiumMind AI — User Journey Maps

Every persona is mapped across stages, with actions, emotional state, pain points and the **AI touchpoint** that StadiumMind introduces at each step. Legend for AI touchpoints: ☐proactive (AI acts first) · ☐reactive (AI answers) · ☐human-approved action.

Persona 1 — Fan (Alex)

Profile. 28, travelled from São Paulo, first time at the venue, speaks Portuguese, has a general-admission ticket and a phone on roaming data.

Goal. Get in, find the seat, enjoy the match, avoid queues, get home.

Stage	Action	Emotion	Pain (today)	StadiumMind AI touchpoint
Leave home/hotel	Checks kickoff, plans transport	Excited	Unclear best route/time	☐ App shows kickoff + recommended departure & transit
Approach venue	Arrives by metro	Anxious	Which gate?	☐ App routes to least-crowded gate for the seat
Security/entry	Queues at turnstile	Impatient	Long unclear queues	☐ Live gate wait times; redirect suggestion if congested
Concourse	Finds the seat	Lost	Poor signage	☐ Turn-by-turn indoor wayfinding, congestion-aware
In-seat	Wants food/washroom	Comfortable	Long queues, misses play	☐ "Least-crowded washroom / shortest food queue near me"
Incident nearby	Sees crowding	Concerned	No guidance	☐ Push alert + safe detour
Full time	Wants to leave	Tired	Egress crush, transport chaos	☐ Best exit + live transport recommendation
Journey home	Boards transit	Relieved	Overcrowded platforms	☐ Ops meters egress; app staggers guidance

Moments that matter: first gate choice, the "where's my seat" moment, and post-match egress. These are where StadiumMind must feel magical.

Persona 2 — Volunteer (Maria)

Profile. 45, local, multilingual, assigned to the west concourse, carries a phone with the volunteer app.

Goal. Help visitors and respond where she's needed, without guessing.

Stage	Action	Emotion	Pain (today)	AI touchpoint
Shift start	Checks in	Ready	Unclear priorities	☐App shows zone status + likely hotspots
Idle	Waits for radio calls	Uncertain	Reactive, radio noise	☐AI suggests proactive positioning near predicted congestion
Dispatch	Receives assignment	Focused	Verbal, error-prone	☐Ops-approved dispatch with turn-by-turn route
On scene	Assists / directs	Purposeful	No situational context	☐Context card: what's happening, what to do
Resolve	Marks complete	Satisfied	Manual logging	☐One-tap status; auto-logged with timestamp

Persona 3 — Operations Manager (Diego)

Profile. 52, runs the operations centre, 20 years of experience, responsible for the whole venue's flow and safety.

Goal. Keep the venue running smoothly without being buried in data.

Stage	Action	Emotion	Pain (today)	AI touchpoint
Pre-match	Reviews readiness	Calm	Manual checks	☐Readiness snapshot; predicted arrival curve
Doors open	Monitors gates	Alert	CCTV wall overload	☐Heatmap + ranked recommendations, not raw alarms
Build-up	Congestion forms at a gate	Stressed	Notices late	☐Prediction: "Gate 3 congested in ~6 min"
Decision	Chooses response	Pressured	No decision support	☐"Redirect to Gate 5 (saves 12 min), 92% conf" → Approve
Incident	Coordinates response	High-stress	Fragmented comms	☐One-tap dispatch + fan reroute; single timeline
Full time	Manages egress	Focused	Transport mismatch	☐Egress metering aligned to live transit capacity
Post-match	Debrief	Reflective	No structured record	☐Auto after-action report from the decision log

Design implication: Diego must feel he has a *copilot*, not another dashboard. Confidence + explanation + one-tap approve is the core loop.

Persona 4 — Security Officer (Sam)

Goal. Detect incidents fast and coordinate a safe, quick response.

Stage	Action	Emotion	Pain (today)	AI touchpoint
Patrol	Monitors zones	Vigilant	Blind spots	☐ Anomaly/crowd-crush risk flag with location
Detect	Confirms incident	Urgent	Slow triage	☐ Guided triage: type, severity, zone
Coordinate	Directs responders	Tense	Radio-only	☐ Fastest clear route; auto fan reroute to protect corridor
Contain	Manages scene	Focused	No live picture	☐ Live density around scene; escalation suggestions
Close	Stands down	Relieved	Manual report	☐ Timestamped audit trail auto-generated

Persona 5 — Medical Team (Dr. Okafor)

Goal. Reach the patient by the fastest clear route and hand off cleanly.

Stage	Action	Emotion	Pain	AI touchpoint
Standby	Awaits calls	Ready	Static positioning	☐ Suggested staging near predicted demand
Alert	Receives medical incident	Focused	Location ambiguity	☐ Exact zone + fastest step-free route
En route	Moves through crowd	Rushed	Crowds block path	☐ Corridor protection: fans rerouted ahead
Treat	Stabilises patient	Intense	No context	☐ Nearest medical point / evac route
Handoff	Transfers care	Careful	Manual logging	☐ Auto-logged incident record

Persona 6 — Accessibility user (Priya, wheelchair user)

Goal. A dignified, fully step-free experience equal to everyone else's.

Stage	Action	Emotion	Pain (today)	AI touchpoint
Arrival	Finds accessible entry	Hopeful	Unclear a11y routes	☐ Accessible gate + step-free route pre-selected
Navigate	Uses lifts/ramps	Cautious	Lifts crowded/ broken	☐ Live lift status; reroute if a lift is down/ busy
Amenities	Reaches accessible WC/food	Independent	Poor info	☐ Nearest accessible amenity with live status
Emergency	Needs safe evac	Fearful	Generic evac ignores a11y	☐ Dedicated step-free evacuation route + volunteer dispatch
Egress	Leaves safely	Confident	Crowded step-free exits	☐ Accessible egress timed to avoid crush

Non-negotiable: accessibility is a *mode across every flow*, not a separate lesser app.

Persona 7 — Transport authority (Metro control)

Goal. Match egress flow to transport capacity to avoid platform crush.

Stage	Action	Emotion	Pain	AI touchpoint
Pre-egress	Forecasts demand	Planning	Guesswork	☐ Predicted egress curve by exit/mode
Full time	Sees surge	Reactive	Platform overcrowding	☐ Recommended metering: hold/release by gate
Coordinate	Aligns with Ops	Collaborative	Siloed comms	☐ Shared plan; Ops approves staggered release

Cross-persona "golden thread" (the demo narrative)

1. Fan is routed to the least-crowded gate (☐).
2. Crowd agent predicts Gate 3 congestion; Ops gets an explained redirect (☐).
3. Volunteers repositioned proactively (☐).
4. Medical incident → responder dispatched, corridor protected, fans rerouted (☐).
5. Post-match: best exit + transport recommended; egress metered with transit (☐).

Every persona experiences the *same event* from their own role — one shared truth, many tailored views.

StadiumMind AI — Information Architecture (IA)

Defines every screen, how screens connect, entry points, deep links and URL maps across all five surfaces: **Fan app**, **Operations**, **Security**, **Medical**, **Volunteer**, plus the shared **Welcome/Language** shell and **Admin**.

1. Product surfaces & top-level map

```
StadiumMind AI
├── Welcome shell (shared entry)
│   ├── Language select
│   └── Role / experience launcher (Fan · Ops · Map · AI Chat)
├── Fan app (mobile) [role: fan]
├── Operations dashboard (web) [role: operations]
├── Security console (web) [role: security]
├── Medical view (web/tablet) [role: medical]
├── Volunteer app (mobile) [role: volunteer]
└── Admin (web) [role: admin]
```

2. Navigation model

- **Fan app:** 5-tab bottom navigation, max **2 taps** to any feature, bottom sheets instead of modals, a floating voice button on every screen.
- **Ops/Security/Medical:** single-screen "cockpit" (no deep menu trees) — map + recommendations + timeline visible together; panels, not page hops.
- **Volunteer:** task-centric single stack (current assignment → detail → status).

Fan bottom navigation (only 5)

```
Home | Map | AI | Alerts | Profile
```

3. Screen inventory & navigation graph

3.1 Welcome shell (shared)

```
[Splash] → [Language Select] → [Experience Launcher]
├── Fan app
├── Ops dashboard
├── Map explorer
└── AI chat
```

Screen	Purpose	Reaches
Splash	Brand + load	Language Select
Language Select	Choose 1 of 6 languages (RTL aware)	Experience Launcher
Experience Launcher	4 cards to enter a surface	Fan / Ops / Map / AI

3.2 Fan app

Home ——— Navigate → Route view → Arrived
 ——— Food → Venue map (food filter) → Bottom sheet → Route view
 ——— Transport → Transport view → Route view
 ——— Washroom → Map (WC filter) → Bottom sheet → Route
 ——— Accessibility → A11y route view
 ——— Emergency → Emergency sheet (call / report / safe exit)

Map ——— Zone tap → Bottom sheet (queue, distance, rating, Navigate)
 AI ——— Suggestions / Voice / Type → Answer card → (Navigate | More)
 Alerts — Alert list → Alert detail → (Navigate to safe zone)
 Profile — Seat info · Language · Accessibility mode · Notifications

Screen	Key elements	Deep link
Home	Match card, 6 action tiles, live alerts	<code>/fan</code>
Map	Google/SVG map, heat overlay, search, bottom sheet	<code>/fan/map</code>
Route view	Path, ETA, crowd density, "Avoid crowd" toggle	<code>/fan/route?to=</code>
AI	Suggestions, voice, chat, action cards	<code>/fan/ai</code>
Alerts	Location-scoped alerts	<code>/fan/alerts</code>
Profile	Seat, language, a11y, notifications	<code>/fan/profile</code>
Emergency	Call, report incident, nearest safe exit	<code>/fan/emergency</code>

3.3 Operations dashboard (single cockpit)

[Top KPI strip] attendance · congested zones · open incidents · match clock
 [Left/Main] Stadium heatmap (zone select → detail)
 [Right] AI recommendations (ranked) → Approve / Reject
 [Bottom] Incident timeline | Volunteers | Transport metering (tabs)

Panel	Purpose	Route
Heatmap	Live congestion by zone	<code>/ops</code>
Recommendations	Ranked, explained, approve/reject	<code>/ops</code> (right rail)
Incident timeline	Chronological incidents & status	<code>/ops#incidents</code>
Volunteers	Allocation & dispatch	<code>/ops#volunteers</code>
Transport	Egress metering vs transit load	<code>/ops#transport</code>

Panel	Purpose	Route
Analytics	Post-match after-action report	/ops/analytics

3.4 Security console

[Map w/ risk flags] → [Incident intake] → [Triage] → [Dispatch + corridor protect] → [Audit log]

Routes: /security, /security/incident/:id.

3.5 Medical view

[Standby staging] → [Incident alert] → [Fastest step-free route] → [On-scene] → [Handoff/log]

Route: /medical, /medical/incident/:id.

3.6 Volunteer app

[My assignment] → [Route] → [Status: accept/en-route/on-scene/resolved] → [Messages]

Routes: /volunteer, /volunteer/task/:id.

3.7 Admin

[Venue config] · [Zones & geo] · [Users & roles] · [Feeds/integrations] · [Feature flags] · [Audit]

Route: /admin/*.

4. URL / route map (canonical)

Surface	Route	Notes
Welcome	/	Language select
Launcher	/launch	4 experience cards
Fan	/fan, /fan/map, /fan/ai, /fan/alerts, /fan/profile	Mobile-first
Ops	/ops, /ops/analytics	Cockpit
Security	/security, /security/incident/:id	RBAC
Medical	/medical, /medical/incident/:id	RBAC
Volunteer	/volunteer, /volunteer/task/:id	Mobile

Surface	Route	Notes
Admin	/admin/*	RBAC (admin)
API	/api/*	Proxied to gateway

5. Entry points & deep links

- **QR on ticket** → /fan?seat=N12-R8 (pre-selects seat context).
- **Push notification** → /fan/alerts or /fan/route?to=<safe_zone> .
- **Staff SSO** → role-appropriate cockpit (/ops , /security , /medical , /volunteer).
- **NFC/beacon at gate** (future) → /fan/map?near=<zone> .

6. State & context that persists across screens

- Selected **language** (and text direction) — global.
- **Seat / ticket context** for the fan — global within Fan app.
- **Accessibility mode** — global; alters routing and layout everywhere.
- **Auth role** — determines available surfaces and API scopes.
- **Live crowd snapshot** — shared read-model powering map, alerts and AI.

7. IA principles applied

- **Shallow, not deep:** operators never navigate more than one level from the cockpit.
- **One truth, many views:** the same crowd/incident model renders per role.
- **Progressive disclosure:** bottom sheets reveal detail on demand, not upfront.
- **Consistent placement:** primary action always bottom-right / bottom-sheet CTA.

StadiumMind AI — Low-Fidelity Wireframes

Box wireframes for every screen with annotations. Notation: [] control/tile, () action button, ===== divider, 🗺 map/heat area, » navigation.

FAN APP

F1 · Language Select (Welcome shell)

```

+-----+
| StadiumMind | Logo (SVG), centered
| Choose your language |
|
| [ English ] [ Español ] 6 tiles, 2 cols
| [ Français ] [ Português ]
| [ العربية ] [ Deutsch ] Arabic tile flips layout to RTL
|
| ( Continue » )
+-----+
```

Notes: large tap targets; selected tile highlighted; remembers choice.

F2 · Experience Launcher

```

+-----+
| StadiumMind AI |
| Choose an experience |
| +-----+ +-----+ |
| | FAN | | OPS | 3-4 cards; icon + title + 1-line desc
| +-----+ +-----+ |
| +-----+ +-----+ |
| | MAP | | AI CHAT |
| +-----+ +-----+ |
+-----+
```

F3 · Home

```

+-----+
| * Good evening, welcome | greeting (SVG sun icon)
| Brazil vs Spain | match card (real data)
| FIFA WC 2026 · MetLife · 7:30 |
| ===== |
| [Navigate][ Food ][Transport] 6 action tiles (SVG icons)
| [Washroom][ Ally ][Emergency]
| ===== |
| Live alerts |
| • Gate 3 busy |
| • Food Court A crowded |
+-----+
```

```
| Home Map AI Alerts Profile | bottom nav (5)
+-----+
```

F4 · Map

```
+-----+
| [search.....] [filter] |
|                               |
|  [grid] Stadium + heat [grid] | Google satellite / SVG fallback
|  [grid] green/amber/red pins [grid] | pins colored by congestion
|                               |
+-----+
| Food Court A |
| Queue 6 min · 120 m · ★4.8 |
| ( Navigate ) |
+-----+
```

bottom sheet on tap

F5 · Route view

```
+-----+
| Walking · 2 min      ( x ) |
|                               |
|  [grid] route polyline [grid] |
|  ● you → ● destination |
|                               |
| Crowd density: [bar chart] |
| [ Avoid crowd ● ON ] |
| ( Start ) ( Alt route ) |
+-----+
```

F6 · AI assistant

```
+-----+
| How can I help? |
| ( [mic] Voice ) ( [type] Type ) |
|-----|
| Suggestions: |
| [Where is my seat?] |
| [Least crowded gate] |
| [Food near me] |
| [Wheelchair route] |
|-----|
| » answer card (title + why + |
|   Navigate action) |
+-----+
```

F7 · Alerts

```
+-----+
| Alerts |
| ● red Gate 3 congested |
| ● amber Food Court A busy |
+-----+
```

```

| ● info Metro minor delay |
| ( Navigate to safe zone ) |
+-----+

```

F8 · Profile

```

+-----+
| Alex Fan |
| Seat: North Stand N12 · Row 8 |
| Language: [English ▼] |
| Accessibility mode [ ○ off ] |
| Notifications [ ● on ] |
+-----+

```

F9 · Emergency sheet

```

+-----+
| Emergency |
| ( Call venue safety ) |
| ( Report incident ) |
| ( Nearest safe exit » ) |
+-----+

```

OPERATIONS DASHBOARD

O1 · Cockpit

```

+-----+
| 42' | Attendance 68,240 | Congested 2 | Open incidents 1 | KPI strip
+-----+
| STADIUM HEATMAP | AI RECOMMENDATIONS |
| zones colored | [crowd · HIGH · 92%] |
| pulsing red hot | Redirect Gate 3 → Gate 5 |
| | "Two buses arrived; G5 is 30%" |
| | ( Approve ) ( Reject ) |
| | [emergency · CRIT · 90%] ... |
+-----+
| Incidents | Volunteers | Transport (tabs) |
| 19:12 CRIT medical @ Medical 1 – 3 volunteers dispatched |
+-----+

```

Annotations: no side menu; approve/reject is the primary loop; recommendations ranked by severity × confidence; clicking a zone filters the right rail.

O2 · Analytics / after-action

```

+-----+
| Match summary · queue time ↓ · incidents · decisions log |
| charts: attendance curve, congestion timeline |
+-----+

```

```
| Recommendations: 24 issued · 18 approved · 6 rejected |
+-----+
```

SECURITY CONSOLE

S1 · Security cockpit

```
+-----+
| 🗺 map + risk flags 🚩 | INTAKE | |
| | | Type [▼] Severity [▼] Zone[▼] |
| | | ( Create incident ) |
| | | TRIAGE / DISPATCH |
| | | Fastest route ✓ Corridor ✓ |
| AUDIT LOG (timestamped) |
+-----+
```

MEDICAL VIEW

M1 · Medical

```
+-----+
| Standby staging: West Concourse (suggested) |
| ALERT: CRIT medical @ Medical 1 |
| ( Accept ) » fastest step-free route 🗺 map 🚩 |
| On-scene ( ✓ ) Handoff ( ✓ ) |
+-----+
```

VOLUNTEER APP

V1 · Assignment

```
+-----+
| Your assignment |
| Assist medical corridor |
| » Medical 1 · 120 m · 2 min |
| ( Accept ) |
| Status: oen-route oon-scene |
| oresolved |
| ----- |
| Messages from Ops |
+-----+
```

ADMIN

A1 · Admin

```
+-----+
| Venue config | Zones & geo | Users & roles | Feeds | Flags |
| 🗺️ editable venue graph, anchor lat/lng, feed status 🗺️ |
+-----+
```

Wireframe → hi-fi mapping

Each low-fi screen maps to a hi-fi component set defined in `04_DESIGN_SYSTEM.md` (Card, ActionTile, BottomSheet, RecommendationCard, Heatmap, KPIStat, TabBar, AlertRow). The standalone HTML shell and the Next.js app implement F1–F9 and O1 directly.

StadiumMind AI — High-Fidelity Design System

A FIFA-quality, accessible, multilingual design system. All values are provided as **design tokens** so they map 1:1 to Tailwind config and CSS variables.

1. Brand & tone

Confident, calm, trustworthy. The UI should feel like a **premium control surface** — spacious, high-contrast, motion used sparingly to draw the eye to what changed. Never flashy during an incident.

2. Color

2.1 Core palette (tokens)

Token	Hex	Use
<code>--navy</code>	<code>#0B1F3A</code>	Primary brand, headers, dark surfaces
<code>--navy-2</code>	<code>#12294C</code>	Gradient partner
<code>--brand (blue)</code>	<code>#2563EB</code>	Primary actions, links, accents
<code>--brand-soft</code>	<code>#DBEAFE</code>	Selected/ghost backgrounds
<code>--surface</code>	<code>#FFFFFF</code>	Cards
<code>--canvas</code>	<code>#F4F6FB</code>	App background
<code>--text</code>	<code>#0F172A</code>	Primary text
<code>--muted</code>	<code>#64748B</code>	Secondary text
<code>--line</code>	<code>#E2E8F0</code>	Borders/dividers

2.2 Status / congestion scale (semantic)

Token	Hex	Meaning
<code>--success (green)</code>	<code>#16A34A</code>	Normal / go / approved
<code>--warning (amber)</code>	<code>#F59E0B</code>	Moderate
<code>--busy (orange)</code>	<code>#EA580C</code>	Busy
<code>--danger (red)</code>	<code>#DC2626</code>	Congested / critical / reject

Congestion heat = green → amber → orange → red (occupancy 0 → 100%). These four are the *only* colors used for the heatmap so the mapping is unambiguous.

2.3 Contrast rules

All text $\geq 4.5:1$ on its background (WCAG AA). Status colors are paired with an icon or label — never color alone — for color-blind accessibility.

3. Typography

- **Family:** Inter (fallback: system-ui). Numerals use tabular figures for KPIs.
- **Scale (px / line-height):**

Token	Size/LH	Weight	Use
display	32 / 38	800	Match title, big KPI
h1	24 / 30	700	Screen title
h2	20 / 26	700	Section
h3	16 / 22	600	Card title
body	15 / 22	400	Paragraph
label	13 / 18	600	Tiles, chips
caption	11 / 14	600 uppercase	Eyebrow/agent tag

RTL: same scale; text-align flips; the layout mirrors (see §9).

4. Spacing, grid & radius

- **8px base grid.** Spacing tokens: 4, 8, 12, 16, 24, 32, 48.
- **Radius:** --radius: 12px (cards, tiles, buttons); pills use 999px.
- **Container:** max-width 1180px (web); fan app frame 380–420px.
- **Layout:** Ops cockpit is a 380px | 1fr split with a KPI strip on top.

5. Elevation & surfaces

Level	Shadow	Use
0	none, 1px --line border	Cards (default, flat + border)
1	0 1px 2px rgba(15,23,42,.06)	Hover
2	0 8px 24px rgba(15,23,42,.12)	Bottom sheets, popovers
Surfaces prefer borders over heavy shadows for a crisp, data-dense feel.		

6. Iconography

- **Library:** Lucide (consistent 2px stroke). **All icons are SVG — never emoji.**
 - Sizes: 16 (inline), 22 (tiles/nav), 28 (feature).
 - Domain glyphs: navigate, utensils, train, accessibility, alert-triangle, map-pin, radio, heart-pulse, leaf (sustainability).
-

7. Components (library)

Component	Anatomy	States
Button	label (+icon); filled / secondary / danger / ghost	default, hover, active, disabled, loading
ActionTile	icon + label; 1/3 width	default, hover, pressed, focus
Card	title (caption) + content; 1px border, 12px radius	—
RecommendationCard	agent tag + severity badge + title + explanation + confidence bar + Approve/Reject	pending, approved, rejected
BottomSheet	handle + title + meta + primary CTA	peek, expanded
Heatmap	SVG/Google map, zone markers colored by level, pulse on congested	live
KPIStat	big number + label on navy tile	—
TabBar	5 items, icon + label, active = brand	active/inactive
AlertRow	level dot + message + optional CTA	info/moderate/busy/congested
Chip / Suggestion	ghost pill, tappable	default, selected
LanguageTile	flag glyph + language name	default, selected (brand border)
Badge (severity)	low/medium/high/critical color-coded	—

7.1 Confidence bar

A 6px track (--line) filled --brand to confidence% . Always paired with a numeric "Confidence NN%" label.

7.2 Severity badge tokens

low = green-tint, medium = amber-tint, high = orange-tint, critical = red-tint. Text uses the darker shade of the same hue.

8. Motion

- **Durations:** 120ms (micro), 200ms (standard), 320ms (sheet).
- **Easing:** `cubic-bezier(0.2, 0, 0, 1)` (decelerate).
- **Purposeful only:** congested zones pulse (1.4s loop) to draw attention; recommendation cards fade+rise on entry. No decorative animation during incidents (reduced-motion respected via `prefers-reduced-motion`).

9. Internationalization & RTL

- 6 launch languages: English, Español, Français, Português, العربية (RTL), Deutsch.
- All copy externalised to a string catalog keyed by locale.
- **RTL (Arabic):** set `dir="rtl"`; layout mirrors (nav, tiles, sheet CTAs, progress fills); icons that imply direction (arrows) are flipped; numerals per locale. Line-length and truncation tested for German (long compounds).

10. Accessibility

- WCAG 2.1 AA. Contrast $\geq 4.5:1$; status never color-only.
- Targets $\geq 44 \times 44$ px; focus rings visible (2px brand outline).
- Full keyboard nav on web consoles; ARIA roles on interactive SVG map markers.
- Screen-reader labels on every icon-only control; live regions announce new alerts.
- **Accessibility mode** increases target size, enables step-free routing and higher-contrast surfaces globally.

11. Design tokens (machine-readable)

```
{
  "color": {
    "navy": "#0B1F3A", "brand": "#2563EB", "brandSoft": "#DBEAFE",
    "success": "#16A34A", "warning": "#F59E0B", "busy": "#EA580C", "danger":
"#DC2626",
    "surface": "#FFFFFF", "canvas": "#F4F6FB", "text": "#0F172A",
    "muted": "#64748B", "line": "#E2E8F0"
  },
  "radius": { "md": 12, "pill": 999 },
  "space": [4, 8, 12, 16, 24, 32, 48],
  "font": { "family": "Inter", "scale": {
    "display": 32, "h1": 24, "h2": 20, "h3": 16, "body": 15, "label": 13, "caption":
11 } },
  "motion": { "micro": 120, "standard": 200, "sheet": 320,
    "easing": "cubic-bezier(0.2,0,0,1)" }
}
```

These tokens are the single source of truth. `tailwind.config.ts` and the standalone HTML shell both derive their theme from them, guaranteeing the fan app, ops dashboard and demo look identical.

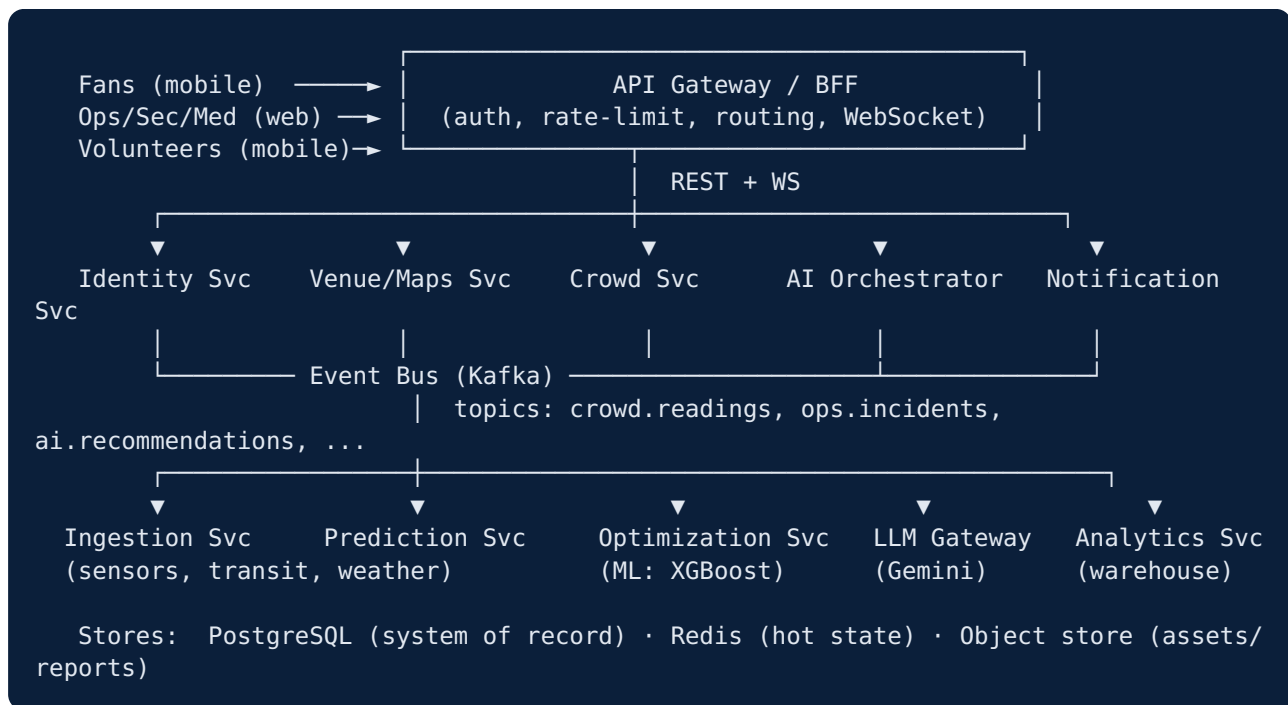
12. Do / Don't

- **Do** lead with the recommended action and its reason.
- **Do** keep one primary CTA per surface.
- **Don't** use more than the four congestion colors on the map.
- **Don't** animate during an active incident beyond the single attention pulse.
- **Don't** use emoji or icon fonts — SVG only.

StadiumMind AI — System Design Document

Covers the target production architecture: microservices, event-driven data platform, AI agent architecture, database schema, API contracts, frontend architecture, security, observability and scaling. The current repository implements a coherent MVP of this design (FastAPI modular monolith with the same service boundaries and graceful fallbacks); this document is the north star it grows into.

1. Architecture overview



Two planes: - **Real-time plane** (event bus + Redis + WebSocket) for live crowd, incidents, recommendations. - **System-of-record plane** (PostgreSQL) for durable entities (incidents, recommendations, users, audit).

2. Microservices

Service	Responsibility	Sync API	Consumes (topics)	Produces (topics)	Store
API Gateway / BFF	AuthN/Z, routing, rate-limit, WebSocket fan-out	all <code>/api/</code> *, <code>/ws/</code> *	ai.recommendations, crowd.readings	—	—
Identity	Users, roles, tokens (OIDC)	<code>/auth/</code> *	—	user.events	Postgres
Venue/Maps		<code>/venue</code> , <code>/directions</code>	—	—	Postgres

Service	Responsibility	Sync API	Consumes (topics)	Produces (topics)	Store
	Venue graph, geo projection, directions proxy				
Ingestion	Normalise sensor/transit/weather feeds	webhooks	—	crowd.readings, transit.updates, weather.updates	—
Crowd	Maintain live density read-model, alerts	/crowd/*	crowd.readings	crowd.alerts	Redis
Prediction	Short-horizon crowd forecasting (ML)	/predict	crowd.readings	crowd.forecasts	model store
Optimization	Queueing, routing, allocation (OR)	/optimize/*	—	—	—
AI Orchestrator	Multi-agent proactive/reactive reasoning	/ai/*	crowd., ops.incidents, transit.	ai.recommendations	Postgres (recs)
LLM Gateway	Gemini access, prompt mgmt, caching, fallback	internal	—	—	cache
Incident	Incident lifecycle & dispatch	/incidents/*	—	ops.incidents, dispatch.commands	Postgres
Notification	Push/SMS/in-app fan & staff messaging	/notify/*	crowd.alerts, ai.recommendations	notify.sent	—
Analytics	After-action, KPIs, warehouse ETL	/analytics/*	all	—	Warehouse

Boundaries follow the current code packages (`core` , `infra` , `simulation` , `ml` , `optimization` , `llm` , `agents` , `api`) — each maps to one or more services when extracted from the MVP monolith. Deployment model: **one logical tenant per venue**, config-driven by the venue model.

3. Event-driven architecture

3.1 Topics (Kafka)

Topic	Key	Payload	Producers → Consumers
crowd.readings	zone_id	occupancy, capacity, arrival_rate, ts	Ingestion → Crowd, Prediction, AI
crowd.forecasts	zone_id	predicted_ratio, minutes_to_congestion	Prediction → AI, Ops
crowd.alerts	zone_id	level, message	Crowd → Notification, Fan/Ops

Topic	Key	Payload	Producers → Consumers
transit.updates	station	mode, load, delay	Ingestion → AI (Transport agent)
weather.updates	venue	temp, precip, wind	Ingestion → AI (Sustainability/Emergency)
ops.incidents	incident_id	type, severity, zone, status	Incident → AI, Security, Medical
ai.recommendations	rec_id	agent, title, explanation, confidence, severity	AI → Ops, Notification
dispatch.commands	volunteer_id	task, route, zone	Incident/AI → Volunteer
notify.sent	user/zone	channel, message	Notification → Analytics

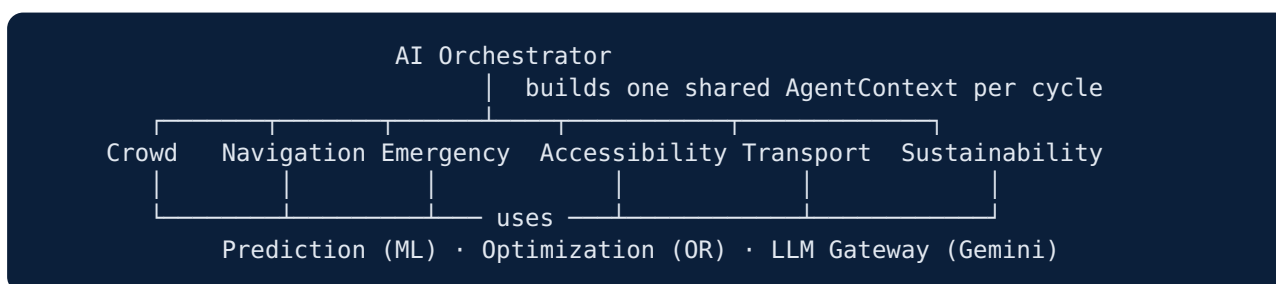
3.2 Patterns

- **Event carried state transfer** for crowd readings (each event self-describes a zone).
- **CQRS**: Crowd service builds a Redis **read-model** (latest snapshot) from the stream; writes (incidents, recommendations, decisions) go to Postgres.
- **Idempotency**: recommendations use **deterministic ids** (`hash(agent|title|zone)`) so re-analysis upserts rather than duplicates and operator decisions persist.
- **Outbox pattern** for reliable publish from services that also write Postgres.
- **Backpressure**: consumers are horizontally scaled by partition (partition key = zone/venue).

3.3 Delivery semantics

At-least-once on the bus; consumers are idempotent. The WebSocket fan-out is best-effort with a periodic full-snapshot to self-heal missed deltas.

4. AI agent architecture



- **Contract**: every agent is an `Agent` subclass implementing `analyze(context)` (proactive) and optionally `handle_query(context)` (reactive), declaring the `intents` it can answer. New capability = new subclass + register (Open/Closed principle).
- **Proactive cycle**: orchestrator builds an `AgentContext` (venue + live crowd + match minute + incidents + tools), fans it to every agent, then **de-duplicates and ranks** recommendations by `severity × confidence`. A failing agent is isolated and logged — it can't break the cycle.
- **Reactive routing**: an intent (e.g. `navigate`, `best_exit`, `wheelchair_route`) is routed to the first capable agent.

- **Explain-before-recommend:** each agent turns its computed situation into prose via the **LLM Gateway** (`explain()`), with a deterministic fallback so an LLM outage never blocks a recommendation.
- **Human-in-the-loop:** every recommendation starts `approved = null` ; operator Approve/Reject is persisted and preserved across cycles.

4.1 Agent responsibilities

Agent	Signals used	Output
Crowd	occupancy, arrival rate, forecast, queueing wait	redirect fans between gates
Navigation	venue graph, live congestion	congestion-aware / step-free routes
Emergency	active incidents, LP allocation	volunteer dispatch + fan reroute
Accessibility	lift/ramp status, a11y graph	step-free routing, a11y risk flags
Transport	transit load, match clock	best egress option, metering
Sustainability	occupancy	energy-saving hints for idle zones

4.2 Models & optimisation

- **Forecasting:** gradient-boosted regression (XGBoost/LightGBM) on `[current_ratio, arrival_rate, match_minute, trend]` ; linear fallback.
- **Queueing:** Erlang-C (M/M/c) wait-time estimates for service points.
- **Routing:** Dijkstra with congestion-weighted edges (+ accessibility filter).
- **Allocation:** integer LP (PuLP) for volunteer assignment; greedy fallback.

5. Database schema (PostgreSQL, system of record)

```
-- Users & auth -----
CREATE TABLE users (
  id          UUID PRIMARY KEY,
  username    TEXT UNIQUE NOT NULL,
  role        TEXT NOT NULL CHECK (role IN
    ('fan','volunteer','operations','security','medical','admin')),
  locale      TEXT NOT NULL DEFAULT 'en',
  created_at  TIMESTAMPTZ NOT NULL DEFAULT now()
);

-- Venue model -----
CREATE TABLE venues (
  id          UUID PRIMARY KEY,
  name        TEXT NOT NULL,
  anchor_lat  DOUBLE PRECISION NOT NULL,
  anchor_lng  DOUBLE PRECISION NOT NULL,
  footprint_m DOUBLE PRECISION NOT NULL DEFAULT 320
);
CREATE TABLE zones (
  id          TEXT PRIMARY KEY,
  venue_id    UUID REFERENCES venues(id),
```

```

name          TEXT NOT NULL,
type          TEXT NOT NULL,                -- gate|seating|food|washroom|...
x             REAL NOT NULL, y REAL NOT NULL,
lat           DOUBLE PRECISION, lng DOUBLE PRECISION,
capacity      INTEGER NOT NULL,
service_points INTEGER NOT NULL DEFAULT 0,
accessible    BOOLEAN NOT NULL DEFAULT true
);
CREATE TABLE zone_edges (                  -- undirected graph (store both directions)
venue_id UUID, a TEXT, b TEXT, PRIMARY KEY (venue_id, a, b)
);

-- Operations -----
CREATE TABLE incidents (
id          TEXT PRIMARY KEY,
venue_id    UUID REFERENCES venues(id),
type        TEXT NOT NULL,                -- medical|security|crowd_crush|fire|
lost_person
severity     TEXT NOT NULL,                -- low|medium|high|critical
zone_id     TEXT REFERENCES zones(id),
description  TEXT NOT NULL,
status       TEXT NOT NULL DEFAULT 'open', -- open|dispatched|resolved
created_at   TIMESTAMPTZ NOT NULL DEFAULT now(),
resolved_at  TIMESTAMPTZ
);
CREATE TABLE recommendations (
id          TEXT PRIMARY KEY,                -- deterministic: hash(agent|title|zone)
venue_id    UUID,
agent        TEXT NOT NULL,
title        TEXT NOT NULL,
explanation   TEXT NOT NULL,
confidence   REAL NOT NULL,
zone_id     TEXT,
severity     TEXT NOT NULL,
actions      JSONB NOT NULL DEFAULT '[]',
metadata     JSONB NOT NULL DEFAULT '{}',
approved     BOOLEAN,                      -- null=pending
created_at   TIMESTAMPTZ NOT NULL DEFAULT now(),
decided_by   UUID REFERENCES users(id),
decided_at   TIMESTAMPTZ
);
CREATE TABLE volunteer_tasks (
id          UUID PRIMARY KEY,
volunteer_id UUID REFERENCES users(id),
incident_id TEXT REFERENCES incidents(id),
zone_id     TEXT REFERENCES zones(id),
status       TEXT NOT NULL DEFAULT 'assigned', -- assigned|accepted|enroute|
onscene|resolved
created_at   TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE TABLE audit_log (
id BIGSERIAL PRIMARY KEY,
actor UUID, action TEXT, entity TEXT, entity_id TEXT,
detail JSONB, at TIMESTAMPTZ NOT NULL DEFAULT now()
);

-- Time-series crowd (partitioned by day; or a TSDB) -----
CREATE TABLE crowd_readings (
venue_id UUID, zone_id TEXT, occupancy INT, capacity INT,

```

```
arrival_rate REAL, ts TIMESTAMPTZ NOT NULL
) PARTITION BY RANGE (ts);
```

Hot state (latest snapshot, per-zone) lives in **Redis** (crowd:{venue}:latest) for sub-second reads; crowd_readings is the durable history for analytics.

6. API contracts (selected)

Base: /api .Auth: Authorization: Bearer <token> ; RBAC per route. Errors use { "detail": "..."} with standard HTTP codes.

6.1 Auth

```
POST /api/auth/login
req { "username": "op", "role": "operations" }
res { "access_token": "...", "token_type": "bearer", "role": "operations" }
```

6.2 Venue & crowd

```
GET /api/venue → { anchor:{name,lat,lng,footprint_m}, zones:
[ {id,name,type,x,y,lat,lng,capacity,neighbours,accessible} ] }
GET /api/crowd/heatmap →
[ { id,name,type,x,y,lat,lng,occupancy,capacity,ratio,level,accessible } ]
POST /api/crowd/tick → heatmap[] (advance simulation one minute)
POST /api/crowd/surge?zone_id=&people= → { status, zone_id, people } (400 if
people<=0)
GET /api/alerts → [ { id,message,zone_id,level } ]
```

6.3 AI

```
GET /api/ai/recommendations →
[ { id,agent,title,explanation,confidence,zone_id,severity,actions[],metadata,approved } ]
GET /api/ai/intents → { intents:
["navigate","best_exit","wheelchair_route", ...] }
POST /api/ai/query req { intent, params:{origin,destination,...} } →
RecommendationOut
POST /api/ai/recommendations/{id}/decision [ops|security] req { approved:bool } →
RecommendationOut
```

6.4 Incidents

```
POST /api/incidents [ops|security|medical] req
{ type,severity,zone_id,description } → IncidentOut
GET /api/incidents?include_resolved= → IncidentOut[]
POST /api/incidents/{id}/resolve [ops|security|medical] → IncidentOut
```


6.5 Match & directions (external-backed)

```
GET /api/match →  
{ id,home_team,away_team,competition,venue,kickoff_utc,status,home_score,away_score,m  
inute,source }  
GET /api/directions?origin=<zone>&destination=<zone>  
→ { origin,destination,distance_m,duration_min,provider,polyline:  
[[lat,lng]],steps[] }
```

6.6 Realtime

```
WS /ws/live → server pushes { match_minute, heatmap[], recommendations[] } per  
tick
```

Full request/response models are the pydantic schemas in `backend/stadiummind/api/schemas.py`; the OpenAPI spec is served at `/docs`.

7. Frontend architecture (Next.js + Tailwind)

- **Framework:** Next.js (App Router) + TypeScript + Tailwind; `output: standalone`.
- **Rendering:** mostly client components for live data; server components for static shell.
- **State/data:** typed API client (`lib/api.ts`) + polling / WebSocket; no global store needed for MVP (local component state + context for locale/auth).
- **i18n:** locale context + string catalog (`lib/i18n.ts`), `dir` toggled for RTL (Arabic).
- **Design tokens → Tailwind:** `tailwind.config.ts` derives colors/radius/fonts from the design-system tokens (§11 of Design System) so brand is centralised.
- **Maps:** `GoogleStadiumMap` loads Maps JS with `NEXT_PUBLIC_GOOGLE_MAPS_API_KEY`, renders congestion markers + Directions polylines; falls back to SVG `StadiumMap`.
- **Routing (see IA §4):** `/` (language) → `/launch` (cards) → `/fan`, `/ops`, `/fan/map`, `/fan/ai`.
- **Component library** mirrors the design system: `Card`, `ActionTile`, `RecommendationCard`, `BottomSheet`, `StadiumMap`, `GoogleStadiumMap`, `KPIStat`, `TabBar`.
- **Delivery:** Dockerized standalone server; static assets via CDN; edge caching for the shell.

8. Security & privacy

- **AuthN:** OIDC/OAuth2 (staff SSO); signed bearer tokens; short TTL + refresh.
- **AuthZ:** RBAC per route (fan/volunteer/operations/security/medical/admin).
- **Transport:** TLS 1.3 everywhere; HSTS; secrets in a vault (never in code).
- **Data:** at-rest encryption; **aggregate-only** crowd analytics (no PII, no face ID); DPIA per host country; audit log of every operator decision.
- **Abuse/limits:** gateway rate-limiting; per-role quotas; WAF on public edges.

9. Observability & operations

- **Metrics** (Prometheus): API latency/error rate, bus lag, recommendation cycle time, LLM latency/cost, WebSocket connections.
- **Tracing** (OpenTelemetry): request → agent → tool spans.
- **Logging**: structured JSON; **per-agent decision log** for audit/debug.
- **Dashboards/alerts** (Grafana): match-day SLOs; on-call rota + runbooks.
- **Health**: `/api/health` reports LLM backend, predictor backend, match/maps providers and venue — for quick match-day readiness checks.

10. Scaling & resilience

- **Horizontal scale** by venue and by Kafka partition (key = zone/venue).
- **Redis** for sub-second live reads; **Postgres** read replicas for consoles.
- **Stateless services** behind the gateway; autoscale on CPU + bus lag.
- **Graceful degradation** (implemented today): Gemini → mock, Kafka → in-memory, Redis → in-memory, SQL → in-memory, Google → SVG/haversine, football-data → fixture. A venue never goes dark because one dependency fails.
- **DR**: multi-AZ; nightly Postgres backups; infra as code for one-command venue bring-up.

11. Mapping to the current repository (MVP → target)

Target service	Today (in repo)
AI Orchestrator + agents	<code>stadiummind/agents/*</code>
Prediction	<code>stadiummind/ml/crowd_predictor.py</code>
Optimization	<code>stadiummind/optimization/*</code>
Crowd/read-model + bus + cache	<code>stadiummind/infra/*</code> , <code>simulation/*</code>
LLM Gateway	<code>stadiummind/llm/adapter.py</code>
Venue/Maps + Match	<code>stadiummind/core/venue.py</code> , <code>integrations/*</code>
Gateway/BFF	<code>stadiummind/api/app.py</code> + <code>service.py</code>
The MVP is a modular monolith with these seams already drawn, so extraction to	
independent services is mechanical rather than a rewrite.	

StadiumMind AI — Implementation Roadmap

Phased delivery across **Frontend**, **Backend**, **AI** and **DevOps**. Each phase ends in a demoable increment and an explicit exit gate. Durations are indicative (2-week sprints).

Phase 0 — Foundations (done in this repo)

Goal: working, tested MVP monolith with the target seams drawn.

Track	Delivered
Backend	OOP agent orchestrator, simulation, ML, OR, FastAPI + WebSocket, graceful fallbacks
AI	6 agents, explain-before-recommend, human-in-the-loop, idempotent recs
Frontend	SVG stadium map, Fan app + Ops dashboard, standalone demo
Integrations	football-data.org match data, Google Maps + Directions, MetLife geo
Quality	97 unit + system tests green; SQL/optional paths verified

Exit gate: end-to-end demo story passes; docs (this package) approved. □

Phase 1 — Product shell & design system (Sprints 1–2)

Goal: FIFA-quality UX foundation.

- **Frontend:** language select (6 langs, RTL) → experience launcher; tokenised Tailwind theme from the design system; component library (Card, ActionTile, RecommendationCard, BottomSheet, KPISat, TabBar, Heatmap); i18n catalog.
- **Backend:** locale-aware responses; `/venue/geo` hardening.
- **AI:** explanation prompt templates per locale.
- **DevOps:** Dockerfile (standalone Next.js) + Compose (api, web, postgres, redis).

Exit gate: all wireframe screens implemented in hi-fi; a11y audit AA on core flows.

Phase 2 — Real-time platform (Sprints 3–4)

Goal: move from simulation to a real event-driven pipeline.

- **Backend:** extract Crowd, Ingestion, Notification services; Kafka topics; Redis read-model (CQRS); outbox pattern; Postgres schema + migrations.
- **AI:** subscribe to `crowd.readings / forecasts`; publish `ai.recommendations`.
- **Frontend:** WebSocket live feed for Ops heatmap + recommendations.
- **DevOps:** Kafka + schema registry; per-partition autoscaling; load test 90k concurrent.

Exit gate: sustained 500 msg/s/venue at p95 < 300 ms; recommendation cycle ≤ 3 s.

Phase 3 — Incident & dispatch (Sprints 5–6)

Goal: full security/medical/volunteer loop.

- **Backend:** Incident service, dispatch commands, audit log, RBAC hardening.
- **AI:** Emergency agent → LP dispatch + corridor protection at scale.
- **Frontend:** Security console, Medical view, Volunteer app (status flow).
- **DevOps:** audit retention, SIEM export, DR runbook.

Exit gate: detection → on-scene ≤ 4 min in tabletop drill; full audit trail.

Phase 4 — Fan scale & offline (Sprints 7–8)

Goal: 90k fans per venue, resilient on congested networks.

- **Frontend:** offline-first caching, push notifications, SMS fallback for critical alerts.
- **Backend:** edge/CDN for shell; BFF caching; rate limits.
- **AI:** async LLM explanations + caching to control latency/cost.
- **DevOps:** multi-AZ, CDN, chaos testing of every fallback path.

Exit gate: app interactive < 2.5 s on 4G at peak; all fallbacks chaos-tested.

Phase 5 — Multi-venue & analytics (Sprints 9–10)

Goal: operate multiple host venues; prove ROI.

- **Backend:** config-driven venue onboarding; per-tenant isolation; Analytics service + warehouse ETL.
- **AI:** guidance-accuracy tracking; per-venue model tuning.
- **Frontend:** after-action report, KPI dashboards, admin console.
- **DevOps:** one-command venue bring-up (IaC); cost dashboards.

Exit gate: 2+ venues live from one control plane; KPI targets met in pilot.

Phase 6 — Advanced (post-tournament backlog)

Digital twin & IoT sensor fusion · BLE/UWB indoor positioning · drone/robotics dispatch · RL crowd control · predictive maintenance · retail personalization · cross-venue "tournament control tower".

Cross-cutting workstreams

Workstream	Ongoing responsibilities
Security & privacy	DPIA per country, pen-tests each phase, secret rotation
Accessibility	WCAG AA gate on every UI PR; assistive-tech testing each phase
Localization	Continuous string QA for 6 languages incl. RTL
Reliability	SLOs, error budgets, match-day on-call, game-day readiness reviews
Data governance	Aggregate-only analytics, retention policy, access reviews

Indicative team (per phase)

- 2–3 Frontend, 3–4 Backend, 1–2 ML/AI, 1–2 DevOps/SRE, 1 Design, 1 PM, 0.5 Security.

Top delivery risks (see PRD §10 for full register)

1. Network saturation at peak → offline-first + CDN + SMS (Phase 4 priority).
2. Data-feed reliability → fallbacks already built; add feed SLAs (Phase 2).
3. Multi-venue complexity → config-driven tenancy + IaC (Phase 5).
4. AI trust/safety → human approval + audit are non-negotiable throughout.

Definition of "match-ready" (v1 launch)

All Phase 1–4 exit gates met + PRD §11 release criteria satisfied: load, security/DPIA, accessibility, runbooks, and 6-language content QA signed off.